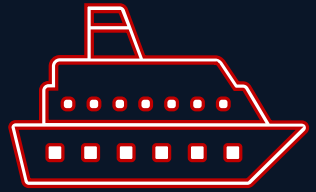
Carnival Cruise Line

MySQL Database Management System
for Ship Operations



ISM 4210 · FINAL PROJECT TEAM REPORT

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BUSINESS BACKGROUND

About Carnival

Is the world's largest cruise line operator
Carries millions of passengers annually
Target market: families, young adults, budget travelers
Ships accommodate thousands at once

- Currently uses manual processes for operations

Business Problems

- Manual work permits & paper-based approvals cause delays
- Paper records risk loss or damage
  - Ex: fires, water spills, emergencies
- Human errors in data entry -> inconsistencies
- Unsustainable usage of paper across voyages

DATABASE DESIGN — ENTITIES & RELATIONSHIPS

Ships

ship_id, name, capacity,
year_built, home_port

Voyages

voyage_id, ship_id, ports,
departure/arrival dates

Crew

crew_id, name, role,
dept, assigned_ship

Tasks

task_id, crew_id, voyage_id,
description, status, time

Passengers

passenger_id, name,
age, loyalty_status

Bookings

booking_id, passenger_id,
voyage_id, cabin, date

Supplies

supply_id, name,
category, unit

Inventory

inventory_id, ship_id,
supply_id, qty, updated

Deliveries

delivery_id, ship_id,
supply_id, qty, date, supplier

DATABASE IMPLEMENTATION

```
SQL --- TABLE CREATION
CREATE TABLE ships (
  ship_id
INT PRIMARY KEY AUTO_INCREMENT,
  name
VARCHAR(100),
  capacity
INT, year_built YEAR,
  home_port
VARCHAR(100)
);

FOREIGN KEY patterns
link all 9 tables via
ship_id, voyage_id,
crew_id, passenger_id,
supply_id
```

9 Tables

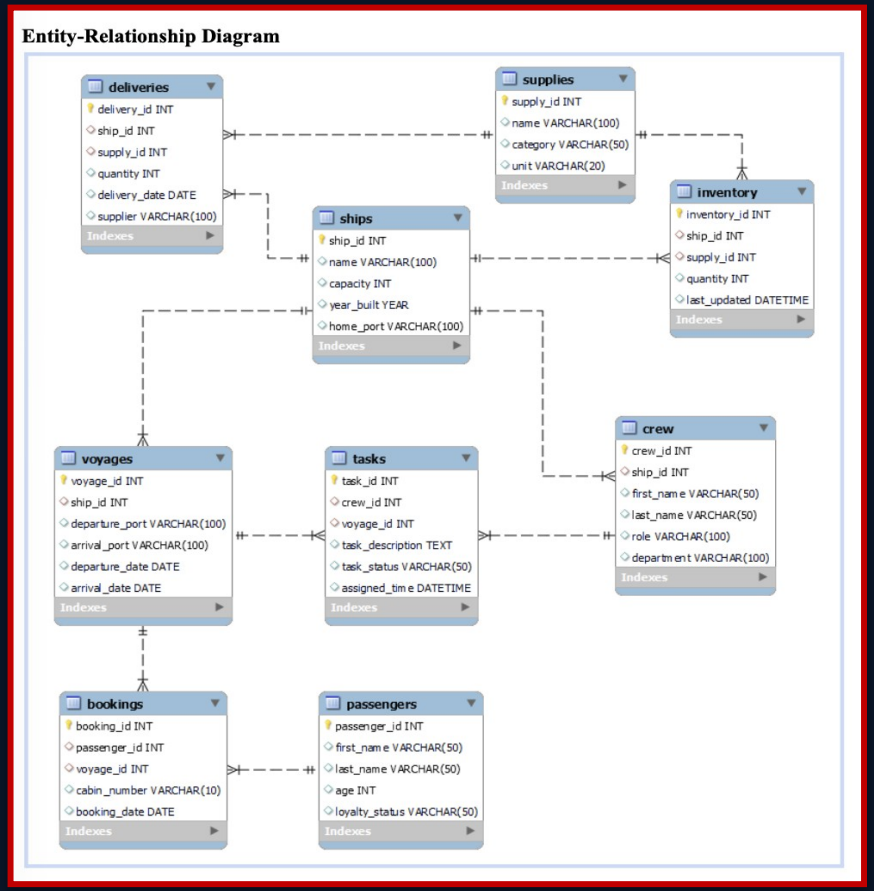
3 Ships

6 Voyages Loaded

19 Crew Members

12 Passengers

ENTITY-RELATIONSHIP DIAGRAM



9 entities · 8 foreign key relationships

SQL QUERY ANALYSIS -- 10 BUSINESS QUESTIONS

Q1 **Unassigned Crew**

LEFT JOIN + IS NULL · Find crew with no tasks assigned

Q2 **Latest Deliveries**

GROUP BY + MAX() · Most recent delivery per ship/supplier

Q3 **Task Overload**

HAVING COUNT > 1 · Identify crew assigned more than one task

Q4 **Loyalty Passengers**

WHERE IN + ORDER BY · Loyalty status of passengers

Q5 **Low Inventory Ship**

SUM + ORDER BY + LIMIT · Ship with lowest quantity of supplies

Q6 **Jamaica Passengers**

JOIN + WHERE date/port · Passengers arriving in Jamaica on June 16th

Q7 **Busiest Voyages**

SUM(CASE) + ORDER BY · Voyages with most in-progress tasks

Q8 **Bahamas Navigator**

JOIN tasks + voyages · Crew member navigating Bahamas route

Q9 **Earliest Pending Task**

MIN(assigned_time) · Oldest unresolved pending task

Q10 **Non-Loyalty Passengers**

WHERE = 'Standard' · Passengers not in loyalty program

SQL QUERY ANALYSIS – Q1

The Staff Manager of Carnival Cruise Line wants to make sure that each employee has been assigned a task. If an employee has not been assigned a task, the Staff manager will look up their role and make sure they are able to be assigned a task as soon as possible.

SQL Code

```
SELECT c.crew_id, c.first_name, c.last_name, c.role
FROM crew c
LEFT JOIN tasks t ON c.crew_id=t.crew_id
WHERE t.crew_id IS NULL;
```

Result

crew_id	first_name	last_name	role
6	Olivia	Grant	Housekeeping Manager
7	Liam	Turner	Bartender
11	Lucas	Brown	Performer
12	Mia	Wilson	Head Chef
13	Logan	Scott	Security Officer
17	Ava	Hall	Housekeeping
18	Elijah	Young	Bartender
19	Harper	Allen	Performer

SQL QUERY ANALYSIS – EXAMPLE 1

The Supply Manager needs to find the ship with the lowest inventory levels. This query helps identify which ship requires immediate restocking.

SQL Code

```
SELECT s.name, SUM(i.quantity) AS total_supplies
FROM ships s
JOIN inventory i ON s.ship_id = i.ship_id
GROUP BY s.ship_id, s.name
ORDER BY total_supplies ASC
LIMIT 1;
```

Result

name	total_supplies
Carnival Horizon	2000

SQL QUERY ANALYSIS – EXAMPLE 2

The HR Manager wants to evaluate operational efficiency by identifying voyages with the most pending or in-progress tasks. This helps prioritize resource allocation.

SQL Code

```
SELECT voyage_id, SUM(task_status = 'in progress') AS in_progress_count, SUM(task_status = 'pending') AS pending_count, COUNT(task_id) AS total_count
FROM tasks
WHERE task_status IN ('in progress', 'pending')
GROUP BY voyage_id
ORDER BY total_count DESC;
```

Result

voyage_id	in_progress_count	pending_count	total_count
2	1	1	2
3	1	1	2
5	2	0	2
1	1	0	1
4	1	0	1
6	0	1	1

SQL QUERY ANALYSIS – EXAMPLE 3

The Director of Fleet Operations was just made aware of a large storm in the Bahamas and wants to notify the crew member responsible for navigating the route. This query helps quickly locate the correct person in urgent situations.

SQL Code

```
SELECT c.first_name, c.last_name, c.role
FROM crew c
JOIN tasks t ON c.crew_id = t.crew_id
JOIN voyages v ON t.voyage_id = v.voyage_id
WHERE t.task_description LIKE 'Navigate Bahamas route';
```

Result

first_name	last_name	role
Maria	Lopez	Captain

SQL QUERY ANALYSIS – EXAMPLE 4

The Marketing Manager wants to identify passengers not enrolled in the loyalty program. This query helps target them with promotions and encourage sign-ups.

SQL Code

```
SELECT passenger_id, first_name, last_name
FROM passengers
WHERE loyalty_status = 'Standard'
ORDER BY first_name DESC;
```

Result

passenger_id	first_name	last_name
11	William	Taylor
8	Olivia	Martinez
6	Emma	Garcia
3	Carlos	Mendez
NULL	NULL	NULL

CONCLUSION + BENEFITS

What this database does for Carnival:

Efficiency

Automates bookings, task assignments, and inventory → eliminating manual paper processes

Safety

Critical crew and passenger data is always accessible for emergency evacuations

Decision Making

10 business queries provide insight into operations, staffing, and supplies

Sustainability

Eliminates paper records across all voyages; reducing environmental impact

Thank You!



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